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MILITARY SPECIFICATION

VALVES; AIRCRAFT HYDRAULIC FLOW REGULATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one class of hydraulic flow regulators suitable for use in Type I aircraft hydraulic systems operating at pressures up to and including 3000 psi. They shall be suitable, however, for use in any hydraulic system with a lower operating pressure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
PPP-B-601	Boxes, Wood Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB, OK 73145-3037 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC F3989 FSC 1650

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Military

MIL-P-116	Preservation, Methods Of
MIL-C-5501	Cap And Plug, Protective, Dust And Moisture Seal
MIL-G-5514	Gland Design, Packings, Hydraulic, General Requirements For
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile And Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base, For Preservation And Operation
MIL-H-8775	Hydraulic System Components, Aircraft And Missiles, General Specification For
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-1523	Age Control Of Age-sensitive Elastomeric Material
MIL-STD-1949	Inspection, Magnetic Particle
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS24391	Plug-Bleeder, Tube, Precision Type
MS28778	Packing, Preformed, Straight Thread Tube Fitting, Boss
MS28886	Valve-Aircraft, Hydraulic Flow Regulator

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI/ASME B46.1-1978	Surface Texture (Surface Roughness, Waviness, and Lay)

(Applications for copies should be addressed to the Society of Mechanical Engineers, 345 East 47th St., New York, NY 10017.)

ASTM B633-85	Standard Specification For Electrodeposited Coating of Zinc on Iron & Steel
ASTM D3951-90	Packaging, Commercial

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(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The requirements of MIL-H-8775 apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

3.2 First article. When specified (see 6.4), minimum and maximum clearance valve samples shall be subjected to first article inspection in accordance with 4.3. A First article inspection test report shall be prepared and provided (see 6.3).

3.3 Design and construction.

3.3.1 General design. The configuration, dimensions, and other details of design shall conform to the requirements of this specification and to the applicable MS standard. The design shall be such as to offer the minimum restriction to flow consistent with the other requirements of this specification.

3.3.1.1 Rate of flow. The flow regulators shall be so designed that they may be calibrated to deliver a constant rate of flow at any increment of inlet pressure ranging up to 3000 psi, and at any increment of outlet pressure up to 2800 psi, or at any combination of the two pressures, provided that the inlet pressure is greater than the outlet pressure plus the maximum allow- able pressure drop (see 6.1).

3.3.2 Plugs. All plugs, except permanently installed plugs that will not have to be removed during the life of the component, shall conform to MS24391 and shall be sealed with gaskets conforming to MS28778. Permanently installed plugs may be pipe-threaded or of any other form suitable for the purpose.

3.3.3 Performance.

3.3.3.1 Rated flow capacity. Rated flow capacity of hydraulic flow regulators shall be in accordance with the requirements as listed in Table I.

3.3.3.1.1 Flow rate tolerance. Tolerances on regulated flow are independent of line size or type and is as follows for normal and high temperatures:

Nominal flow below 0.49 gpm	+20 percent, -20 percent
Nominal flow 0.5 to 1.49 gpm	+10 percent, - 15 percent

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Nominal flow above 1.5 gpm +5 percent, - 10 percent

Tolerances for minus 54°C (minus 65°F) tests shall be ±20 percent for all rates of flow.

TABLE I. Rated flow capacity.

MS Part No.	Tube Size (Inches)	First Dash No.	Individual Flow Settings-gpm (Flow rate in gpm is second Dash number of MS Part No.)						
MS28886-	1\4	4	0.12	0.25	0.38	0.50	0.75	1.0	1.2
MS28886-	3\8	6	1.5	1.75	2.0	2.5	3.0	3.5	-
MS28886-	1\2	8	4.0	4.5	5.0	5.5	6.0	-	-
MS28886-	5\8	10	6.5	7.5	8.5	9.5	10.5	-	-
MS28886-	3\4	12	12.0	13.0	14.0	15.0	16.0	-	-
MS28886-	1	16	17.0	20.0	23.0	26.0	29.0	-	-

3.3.3.2 Damping. Hydraulic flow regulators shall be independent of any normally encountered back-pressures, or pressure surges which might tend to cause them to operate or bypass fluid in a manner other than intended or desired. Opening or closing of the hydraulic flow regulator shall not, in any conceivable normal operation, create shock waves or pressure surges.

3.3.4 Detail drawing. When specified in the contract or order, assembly and detail drawings shall be prepared and provided (see 6.3).

3.4 Finish.

3.4.1 Surface roughness. Surface roughness finish shall be established and shall be specified in manufacturer's drawings as outlined in ANSI/ASME B.46.1-1985. The determination of surface finish shall be made by profilometer, comparator brush analyzer, or other suitable comparison equipment with an accuracy of ± 5 micro inches at the level being measured.

3.4.2 Steel or copper-alloy parts. Unless otherwise specified, all steel or copper-alloy parts, except those in constant contact with hydraulic fluid during use or storage, shall be coated with ion vapor deposited aluminum in accordance with MIL-C-83488, Type I or II, zinc-plated in accordance with ASTM B633-85, or chrome-plated in accordance with QQ-C-320. Surfaces in sliding contact with each other may be chrome-plated.

3.4.3 Other finishes and paints. No finishes or paints, other than those specified above, or color markings specified herein or otherwise authorized by the procuring activity shall be applied to the components either externally or internally prior to installation on the aircraft.

3.5 Sub-zero stabilization of steel. All close-fitting steel parts shall be subjected to sub-zero stabilization to reduce warpage tendencies. The stabilization process shall be conducted as follows:

a. Cool the part to minus 84 ± 6°C (minus 120 ± 10°F).

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b. Hold the above temperature for one hour per inch of maximum section thickness of the part.

c. Heat to 240°C (375°F) and hold for one hour per inch of maximum section thickness of the part.

3.6 Magnetic particle inspection. All magnetizable, highly stressed parts shall be subjected to magnetic particle inspection. Where necessary, these inspections shall be called for on the manufacturer's drawings.

3.7 Age controls. Age-sensitive elastomeric parts shall be in accordance with MIL-STD-1523.

3.8 Recovered materials. Recovered materials are used to the maximum extent possible without jeopardizing the end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of know defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified here are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. All first article inspections shall be performed on the maximum sample valves with the exception of the extreme temperature operation and the immersion tests. The minimum sample valves shall be subjected to these tests.

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4.3.1 First article inspections.

a. Quality conformance inspections and tests (see paragraph 4.4)
b. Immersion (see paragraph 4.3.1.1)
c. Pressure drop (see paragraph 4.3.1.2)
d. Pressure differential compensation (see paragraph 4.3.1.3)
e. Operating low range (see paragraph 4.3.1.4)
f. Emergency operation (see paragraph 4.3.1.5)
g. Extreme temperature operation (see paragraph 4.3.1.6)
h. Endurance (see paragraph 4.3.1.8)
i. Burst pressure (see paragraph 4.3.1.9)
j. Impulse (see paragraph 4.3.1.8)

4.3.1.1 Immersion tests.

4.3.1.1.1 Water immersion plastic parts. One minimum clearance valve sample from each of the compoundings of plastic material shall be immersed in water 25 ± 2°C (77° ± 4°F) until maximum rate of dimensional change does not exceed 0.0003 inch in a 10 day period, as shown by a faired curve plot of the measurements. This test shall cover a period of not less than 30 days. Dimensional measurements shall be made daily for the first 14 days, and at least every 2 days for the remainder of the period. If the unit is made of several plastic parts, of which some remain stationary during operation of the unit, the immersion test should be conducted in such a manner that the swelling action will produce the worst possible conditions towards malfunctioning of the unit. For example, if the unit employs a plastic poppet sliding within a stationary plastic guide which is retained by a metal housing, the swelling action of the guide may be restricted by the metal housing. In such a case, the guide should be assembled in the mating bore during immersion to determine if the swelling action will reduce the clearance between the guide and the poppet, and if excessive strains are set up in the guide. The measurements made shall prove that there will be no interference between mating parts which would impair operation. If the manufacturer has obtained a waiver of the first article inspection of this specification, the minimum clearance sample shall be fabricated to the minimum clearance dimension, plus the allowance for maximum swell after immersion.

4.3.1.1.2 Oil immersion plastic parts. One minimum clearance valve sample from each of the compoundings of the plastic material shall be immersed in hydraulic fluid conforming to MIL-H-5606 at 70 ± 1°C (158 ± 2°F) for a period of 30 days. This oil immersion shall be subject to the same conditions as specified in 4.3.1.1.1 for water immersion, and measurements shall be made at similar periods of immersion.

4.3.1.1.3 Other non-metallic parts. Hydraulic flow regulators containing non-metallic parts other than plastic parts or AN standard seals in glands conforming to MIL-G-5514, shall be immersed in hydraulic fluid for a period of seven days at a temperature of 70 ± 1°C (158 ±2°F) prior to conducting the first article inspection specified herein. All internal parts shall be in contact with the fluid during this period. After the above immersion, the hydraulic flow regulator shall remain in the fluid at normal room temperature until ready for test. The flow regulator shall not be exposed to the air for any appreciable length of time during the immersion test.

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4.3.1.2 Pressure drop. The pressure drop of the flow regulator shall not exceed 70 psi at 21°C to 43°C (70°F to 110°F) or 200 psi at minus 54°C (minus 65°F), at any flow rate up to the calibrated flow rate minus the tolerance allowed on Table I, when tested with slowly increasing flow applied in the normal direction and repeated in the reverse direction.

4.3.1.3 Pressure differential compensation. With inlet pressure of 200 psi, and a flow passing through the regulator, the regulator shall react to a sudden increase of inlet line pressure resulting from the discharge through a quick-opening valve of a 400 cubic inch (min) accumulator charged to 2800 psi, and the momentary overshoot of flow rate shall not exceed 20 percent of the nominal flow rate and such momentary overshoot shall not exceed 1/2 second duration. With inlet pressure at 3000 psi and outlet pressure of 200 psi the regulator shall react to a sudden increase of outlet pressure to 2800 psi, and the momentary effect of such reduced differential shall not cause the flow rate to drop below 75 percent of the nominal flow rate of the regulator being tested, nor shall such momentary reduction exceed 1/2 second duration.

4.3.1.4 Operating flow range. The test apparatus shall be arranged substantially as shown in figure 1. Hydraulic pressure shall be supplied by means of a constant displacement, power-driven pump of a capacity exceeding the maximum permissible flow rate of the regulator being tested. Provisions shall be made to apply pressure suddenly with a charged accumulator in the circuit or gradually (with the accumulator valve off) by closing the bypass valve. The relief valve shall be adjusted to 3000 psi. The operating flow range test shall be conducted by closing the accumulator valve and opening the quick-opening valve and the loading valve. The bypass valve shall then be adjusted until the inlet pressure of the flow regulator is 70 psi greater than the outlet pressure (the outlet pressure will depend only on the configuration of the return line). The regulated flow rate obtained for this condition shall be within the limits specified in Table I. By adjusting the bypass valve slowly, the inlet port pressure shall be increased and the regulated flow rate noted at approximately 10 equal increments of pressure ranging up to 4000 psi. With an inlet pressure of 3000 psi, the loading valve shall be slowly adjusted so as to increase the outlet port pressure and the regulated flow rate obtained shall be noted at approximately 10 equal increments of pressure ranging up to 2900 psi. The regulated flow rate of each increment of inlet and outlet port pressure increases shall be within the limits specified in Table I. The pressure shall then be reduced by opening the bypass valve and the loading valve. The accumulator valve shall then be opened. The pressure shall then be built up in the accumulator by closing the quick-opening valve and the bypass valve. After a sufficient delay to ensure complete accumulator charging, the quick-opening valve shall be opened and flow shall be noted. The test shall be repeated at 2500 psi, 2000 psi, and 1500 psi relief valve settings and regulated flow rate shall be within the limits specified in Table I.

4.3.1.5 Emergency operation. With the moving parts of the flow regulator in their maximum restricted position, and with a 3000 psi pressure differential across the regulator, the flow rate shall be at least 50 percent of the nominal rating of the regulator.

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4.3.1.6 Extreme temperature operation.

4.3.1.6.1 High temperature. The regulator shall be installed in a circuit substantially as shown if figure 2; the closed portion of the circuit shall be bled and the temperature of the closed circuit and regulator shall be maintained at 71°C (160°F) minimum for a period of at least six hours. Five cycles each shall then be accomplished at regulator inlet pressures of 500, 1000, 1500, 2500, and 3000 psi. The flow rate during the test shall be within the range specified in Table I.

4.3.1.6.2 Low temperature. The low temperature test shall be conducted in a manner similar to the high temperature test, except that prior to testing the closed portion of the circuit, the regulator shall be maintained unpressurized at a temperature of not warmer than minus 54°C (minus 65°F) for a period
of four hours and then pressurized to 3000 psi for an additional four hours at minus 54°C (minus 65°F). The regulated flow rate shall not exceed the limits as specified in Table I.

4.3.1.6.3 Rapid warm up. The circuit used in the low temperature test shall be allowed to warm up to plus 22°C (plus 40°F) as rapidly as room temperature will allow. With the relief valve set at 1500 psi at regulator inlet port, at least six cycles of operation shall be imposed at regular intervals during warm up. The regulated flow rate shall be within the limits specified in Table I.

4.3.1.7 Endurance. Pressure at 3000 psi shall be supplied in a manner shown in figure 3. The selector valve shall be cycled at a rate of 35 $\pm$ 5 cycles per minute and the test shall consist of 50,000 cycles of flow regulation and flow reversal. The regulator shall be removed from this circuit after each 10,000 cycles and subjected to the operating flow range test (4.4.3) and shall meet the requirements of those tests. The flow regulator, after completion of the 50,000 cycles, shall be disassembled and examined. There shall be no evidence of deterioration other than normal wear.

4.3.1.8 Impulse. The regulator shall be impulse-tested in a circuit similar to that shown in figure 4. The pressure shall be adjusted to 3000 psi, and the momentary surge pressure shall be adjusted to 4500 psi, and cycling shall be accomplished at a rate of 35 $\pm$ 5 cycles per minute. After 10,000 cycles of operation, the peak pressures may be reduced to 3750 psi, and 40,000 additional cycles shall be imposed. There shall be no evidence of external leakage, permanent set, or other malfunction during the test. NOTE: At the option of the manufacturer, the impulse test may be conducted on a different regulator than the one which was subjected to the testing specified in 4.3.1.7. If a different regulator is used for impulse test, the regulator, after completion of impulse testing, shall pass the test specified in 4.3.1.9.

4.3.1.9 Burst pressure. The flow regulator shall withstand an application of 7500 psi hydrostatic pressure without external leakage or failure. Pressure shall be applied in a manner outlined in 4.4.2 and the rate of pressure rise shall not exceed 35,000 psi per minute.

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4.4 Qualify conformance inspections. All quality conformance inspections shall be performed on all valves, i.e. maximum and minimum valve samples and all of the valves provided to the government. Quality conformance inspections are as follows:

a. Examination of product (see 4.4.1)
b. Proof pressure and leakage (see 4.4.2)
c. Operating flow range (see 4.4.3)

4.4.1 Examination of product. Each hydraulic flow regulator shall be carefully examined to determine conformance to the requirements of this specification for design, workmanship, marking, conformance to the applicable standard and manufacturer's drawings, and for any visible defects. Magnetic particle inspection shall be in accordance with MIL-I-1949. Cracks and other injurious defects disclosed by magnetic inspection shall result in rejection of the component.

4.4.2 Proof pressure and leakage. Hydraulic pressure of 4500 psi shall be applied to the inlet port with the outlet port plugged. Pressure shall be maintained for a minimum of three minutes. There shall be no evidence of external leakage, failure, distortion, or permanent set. With the outlet port plugged, the flow regulator shall withstand a hydrostatic pressure of five psi and 4500 psi. There shall be no external leakage at any of these applied pressures.

4.4.3 Operating flow range. The test apparatus shall be arranged substantially as shown in figure 1. Hydraulic pressure shall be supplied by means of a constant displacement, power-driven pump of a capacity exceeding the maximum permissible flow rate of the regulator being tested. Provisions shall be made to apply pressure suddenly with a charged accumulator in the circuit or gradually (with the accumulator valve off) by closing the bypass valve. The relief valve shall be adjusted to 3000 psi. The accumulator shall be removed or isolated from the circuit. The bypass valve shall then be closed until the inlet pressure is 70 psi greater than the outlet pressure. The flow rate shall be within the requirements specified in Table I. The bypass valve shall then be closed slowly, and as the pressure is increased, the flow rate shall be observed and shall remain within the requirements specified in Table I up to a differential pressure of 3000 psi.

4.5 Test conditions. This change is required because th existing paragraph is in conflict with other paragraphs of this specification.

4.5.1 Hydraulic fluid. First article tests shall be performed with fluid conforming to MIL-H-5606 or MIL-H-6083.

4.5.1.1 Filtration. The test fluid shall be continuously filtered through a filter element which is equivalent to a 25-micron absolute standard filter element. The filter and element used shall be satisfactory for the temperature range encountered, and shall be cleaned or changed regularly to avoid clogging.

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4.5.2 Temperature. Except where otherwise specified, the tests of this specification shall be conducted at a room temperature between 21°C and 32°C (70°F and 90°F) and an oil temperature between 21°C and 43°C (70°F and 110°F). The actual temperature shall be recorded.

4.5.3 Sequence of tests. All valves shall be inspected and tested in the sequence shown in paragraph 4.4 - Quality conformance inspections. The first article valves shall be inspected and tested in the sequence shown in paragraph 4.3 - First article inspections.

4.5.4 Cleaning. All grease or foreign material shall be removed from the interior parts of the valve before testing.

4.5.5 Instruments. Calibrated instruments shall be used for all measurements.

4.5.6 Air removal. Air shall be completely removed from all hydraulic portions of the flow regulators prior to first article and quality conformance inspections.

4.5.7 First article inspections. The specified valves shall have met inspections and successfully performed in accordance with first article inspections specified herein.

4.5.8 Quality conformance. All valves offered to the government shall have met inspections and successfully performed in accordance with quality conformance inspections.

4.5.9 Internal treatment and closure. Following the completion of inspections and tests specified herein, each flow regulator shall be internally flushed with hydraulic preservative oil conforming to MIL-H-6083. All ports shall be capped or plugged with closures conforming to MIL-C-5501 and, as required, prepared for delivery in accordance with Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C, or Industrial IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Flow Regulator shall be cleaned in accordance with (IAW) process C-1 of MIL-P-116.

5.1.1.2 Drying. Flow Regulator shall be dried IAW process D-1 of MIL-P-116.

5.1.1.3 Preservative application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each flow regulator shall be packaged in weather resistant, with

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sufficient cushioning material between bag and unit container of a type, density, and thickness to ensure shock transmission does not exceed peak values in G's established for the flow regulator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each flow regulator shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Commercial. The commercial preservation of the flow regulator shall be IAW ASTM D3951-90.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Flow regulator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare, and cube consistent with the protection required.

5.2.2 Level B. Flow regulator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged flow regulator shall be packed IAW ASTM D3951-90.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked IAW MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hydraulic flow regulator valves covered by this specification are intended for use in aircraft hydraulic systems as specified in MIL-H-5440 to control the rate of flow by inducing a precise degree of pressure drop where reverse flow in excess of regulated flow is not anticipated, or in applications where high pressure drops during reverse flow is not objectionable.

6.1.1 Surging conditions. Caution should be observed in using hydraulic flow regulators to control hydraulic motors in view of the surging condition that might occur.

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6.1.2 Flow regulators. Flow regulators meeting the requirements of this specification are for use in hydraulic systems containing only hydraulic oil conforming to MIL-H-5606 and not qualified for use with other types of fluid, unless approved by the procuring activity.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Specification sheet part numbers.
c. Applicable levels of preservation, packaging, and packing (see Section 5).
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Data requirements. When this specification is used in acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraphs NO.	Data requirement title	Applicable DID NO.	Option
3.2	Test/Inspection Reports	DI-NDTI-80809A	
3.3.4	Engineering Drawings	DIDRPR-80651	—

(DID's related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5010.12-L AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, sample units shall be inspected (see 3.2, 4.2, and 4.3).

6.5 Subject term (key word) listing.

Fluid Flow Control
Operating Flow Range
Rated Capacity
Type I Component

6.6 Assembly and detail drawings. Two sets of assembly and detail drawings shall be furnished with each valve submitted for first article tests. Assembly drawings shall show a cutaway section of details in their normal assembly position and shall carry part numbers of all details and subassemblies. The following data shall be furnished on or together with all assembly drawings:

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a. Mounting dimensions
b. Port dimensions
c. Over-all dimensions
d. Rated flow capacity
e. Dry Weight
f. Maximum operating pressure
g. Complete dismantling procedure and description of tools needed
h. Any special installation or operating instructions considered necessary

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force -99
Army -AV
Navy -AS

Preparing activity:
Air Force -71

Project number:
1650-0502

Reviewer:
Defense Supply Center -CS

User: Army -MI

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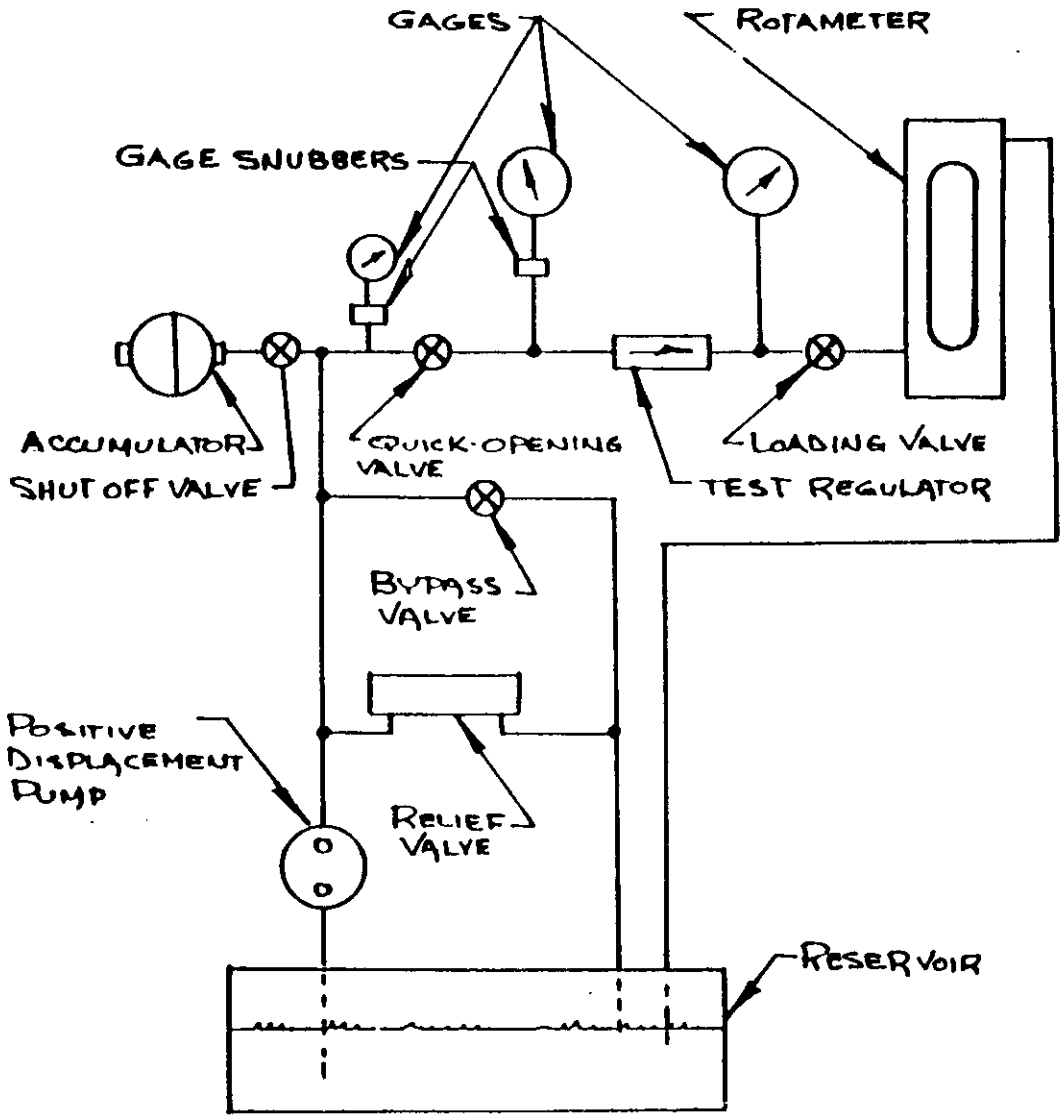


FIGURE 1. Operating flow range test

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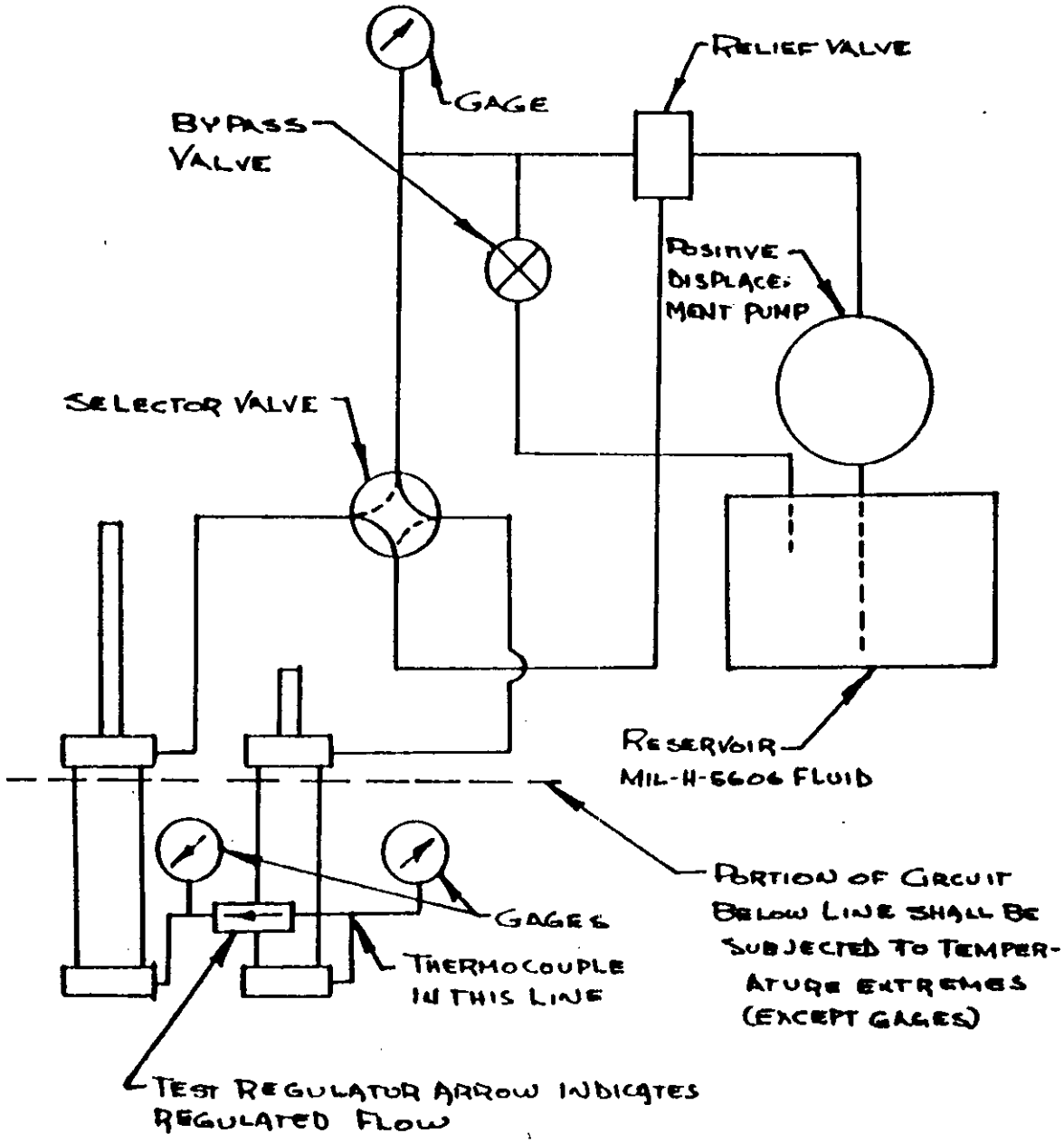


FIGURE 2. Extreme temperature test

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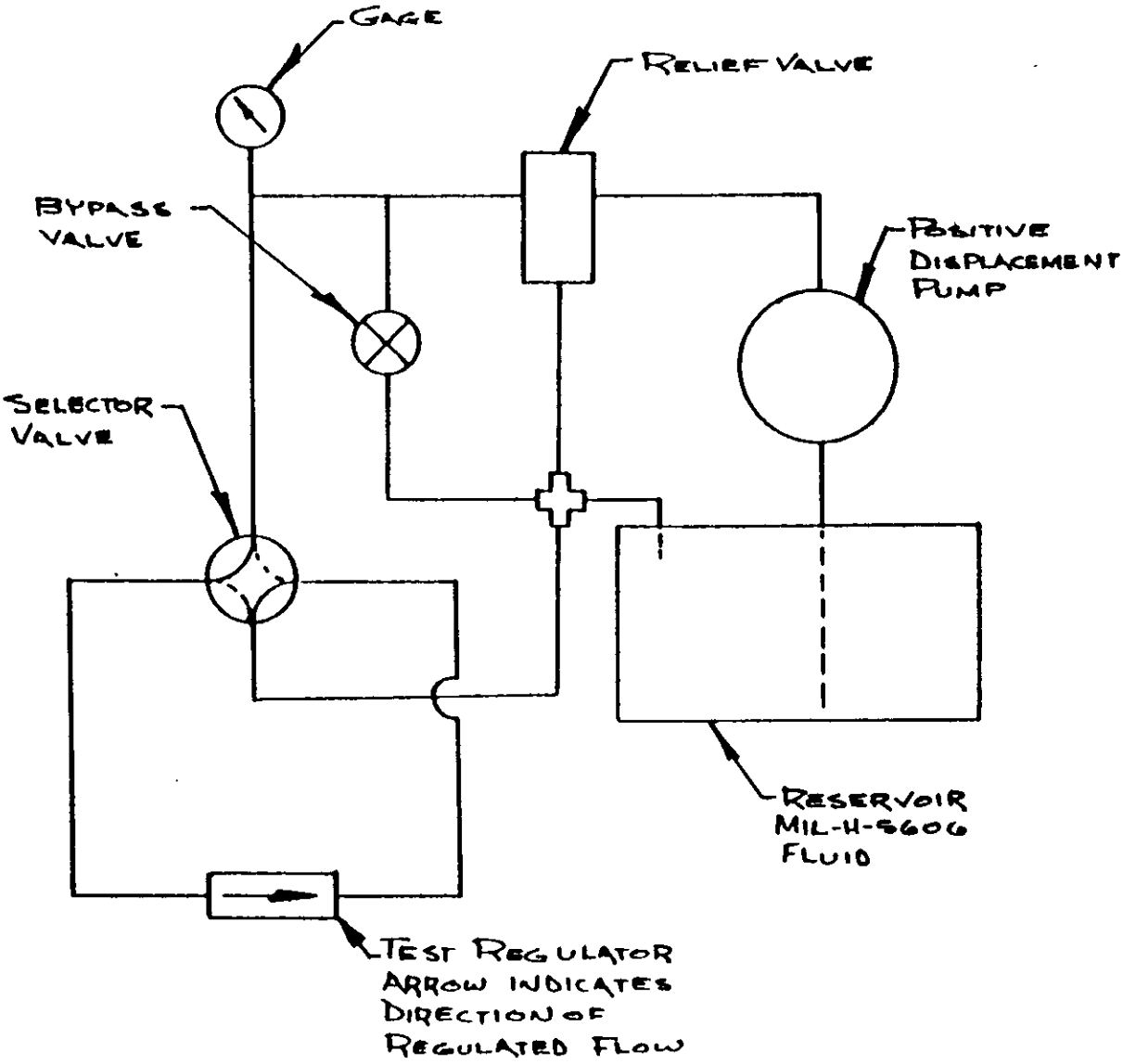


FIGURE 3. Endurance test

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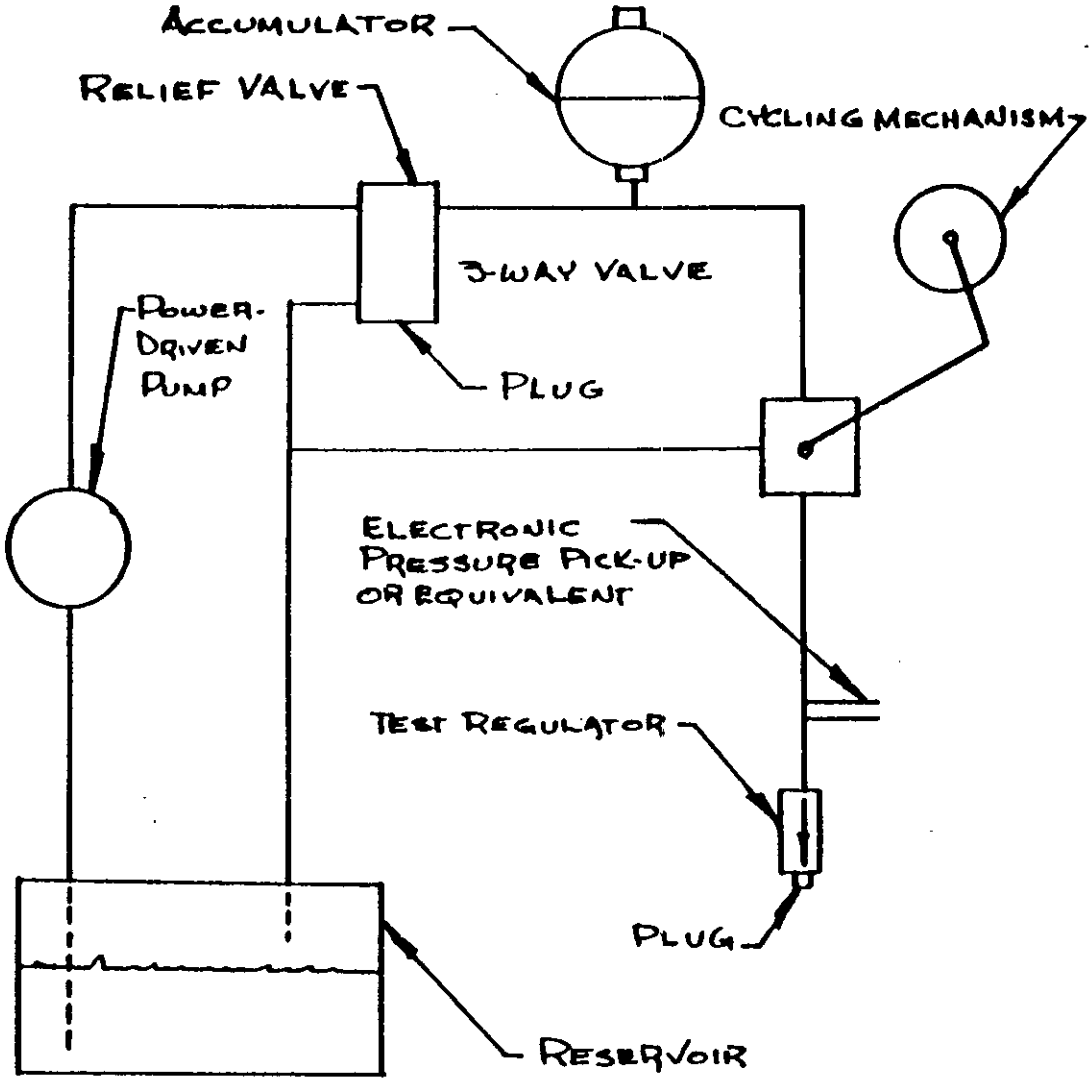


FIGURE 4. Impulse test set-up

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